

of program that displays text. You can press CTRL-F to bring up a Find window to enter a word to find in the program. Often the F3 key will repeat this search to highlight the next occurrence of the word. This feature can save you an extraordinary amount of time compared to manually scrolling through your document to find a word.

Editors also have a find-and-replace feature, which is often assigned the CTRL-H or COMMAND-H shortcut. This allows you to locate occurrences of one bit of text and replace it with another. This is useful if you want to rename a variable or function. However, you need to be careful using the find-and-replace feature because you could unintentionally replace text that matched your find criteria by coincidence.

The Debugger

A debugger is a tool that runs programs one line at a time and lets you inspect the current state of the program's variables. It's a valuable tool for tracking down bugs. This section will explain the features of the Mu Editor's debugger. Don't worry; every debugger will have these same features, even if the user interface looks different.

To start a program in the debugger, use the Debug menu item in your IDE instead of the Run menu item. The debugger will start in a paused state on the first line of your program. All debuggers have the following buttons: Continue, Step In, Step Over, Step Out, and Stop.

Clicking the Continue button causes the program to execute normally until it terminates or reaches a breakpoint. (I describe breakpoints later in this section.) If you are done debugging and want the program to continue normally, click the Continue button.

Clicking the Step In button causes the debugger to execute the next line of code and then pause again. If the next line of code is a function call, the debugger will "step into" that function and jump to the first line of code of that function.

Clicking the Step Over button executes the next line of code, similar to the Step In button. However, if the next line of code is a function call, the Step Over button will "step over" the code in the function. The function's code executes at full speed, and the debugger will pause as soon as the function call returns. Using the Step Over button is more common than using the Step In button.

Clicking the Step Out button causes the debugger to execute lines of code at full speed until it returns from the current function. If you have stepped into a function call with the Step In button and now simply want to keep executing instructions until you get back out, click the Step Out button to "step out" of the current function call.

If you want to stop debugging entirely and not bother to continue executing the rest of the program, click the Stop button. The Stop button immediately terminates the program.

You can set a *breakpoint* on a particular line and let the program run at normal speed until it reaches the breakpoint line. At that point, the